C / System Verilog

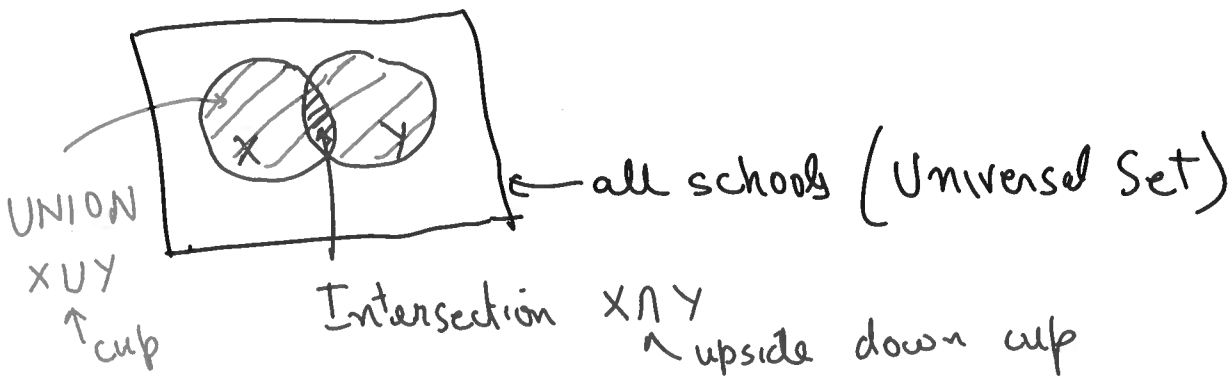
	bitwise	logical
AND	$x \& y$	$x \&\& y$
OR	$x \mid y$	$x \mid\mid y$
NOT	$\sim x$ (tilde)	$! x$

Venn Diagram → Karnaugh Maps = (Truth table + Venn Diagram)

X is a top school

$x \in X$ is a set of all top schools → x is a top school

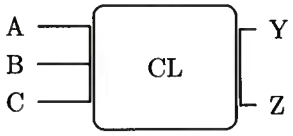
$y \in Y$ is a set of all top sports teams → y has a top sports team

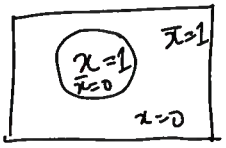


Intersections correspond to AND operations

Unions correspond to OR operations

1.3 Digital circuits or networks

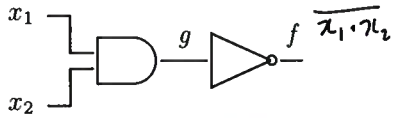


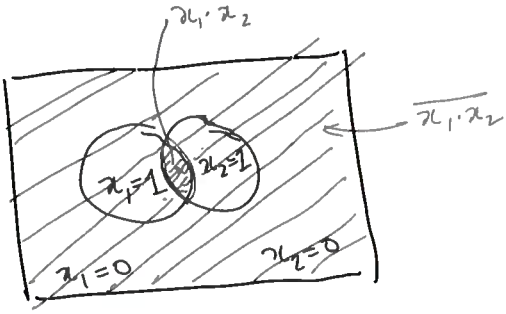


$$Y = F(A, B, C) \qquad Z = G(A, B, C)$$

1.4 Two input networks

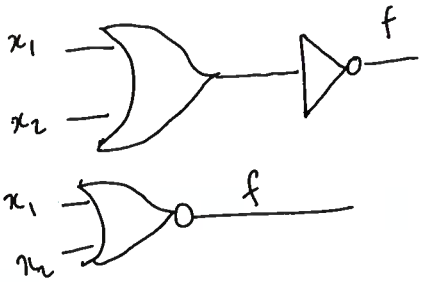
Example 1.2. *Convert the following (ANSI) network into a Boolean expression, a truth table and a Venn diagram.*



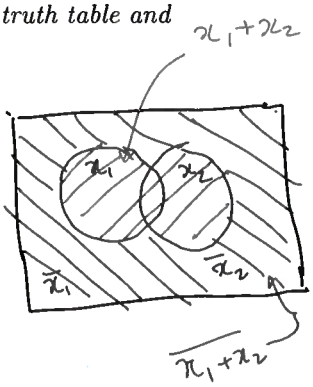


Example 1.3. *Convert the following Boolean expression into a (ANSI) network, a truth table and a Venn diagram:*

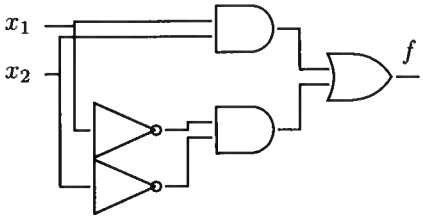
$$f = \overline{x_1 + x_2}$$



x_1	x_2	$x_1 + x_2$	$\overline{x_1+x_2}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

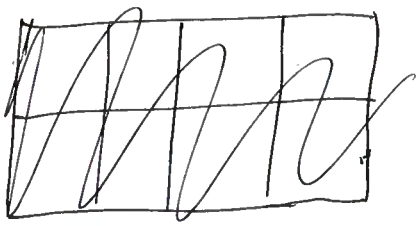


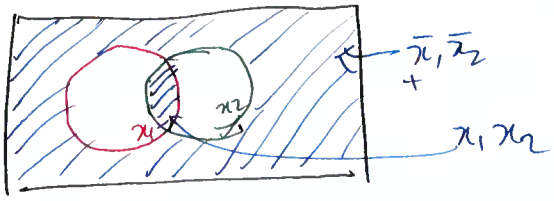
Problem 1.1 (10 marks). *Convert the following (ANSI) network into a Boolean expression, a truth table and a Venn diagram.*



$f = x_1 x_2 + \bar{x}_1 \bar{x}_2$

x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	1

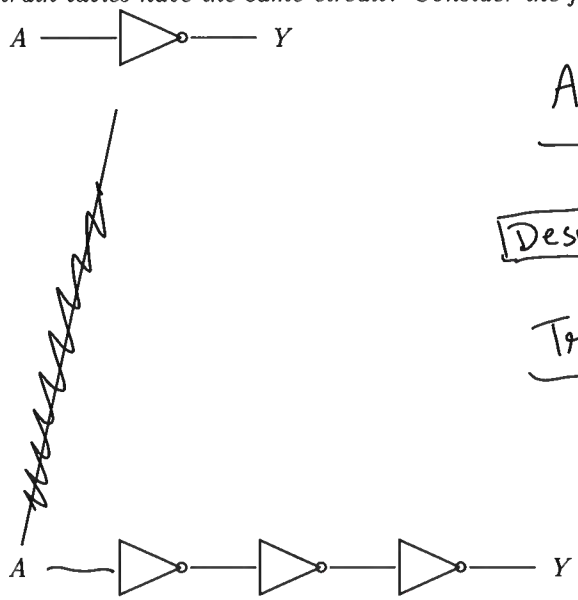




Example 1.4. *Convert the following Boolean expression into a network, a truth table and a Venn diagram:*

$$f = x_1\bar{x}_2 + \bar{x}_1 x_2$$

Problem 1.2 (5 marks). *Can two different circuits have the same truth table? Can two different truth tables have the same circuit? Consider the following two circuits for example*



ANSI networks / Boolean expressions

How

Design

Analysis

Truth table / Venn Diagram

what

Output / Input relationship

~~Answer 1.2 No, two different circuits have the same truth table~~ cannot

Answer 1.2 Yes, two different circuits have the same truth table

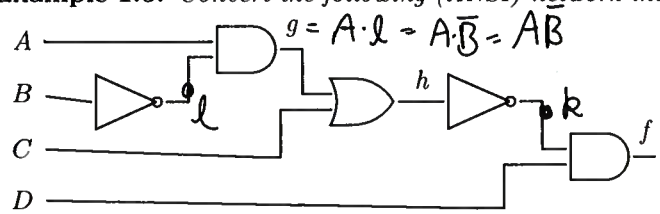
No, two different truth tables cannot have the same circuit

Remark 1.1. *Truth tables and Venn diagrams define* what *the combinational circuit should do. Truth tables define output for every input. Boolean expression and networks define* how *to achieve the desired input output relationship.*

1.5 Multi-input networks

Example 1.5. *Convert the following (ANSI) network into a Boolean expression and a truth table.*

4 inputs



one output

$l = \bar{B}$

$g = A\bar{B}$

$h = g + C = A\bar{B} + C$

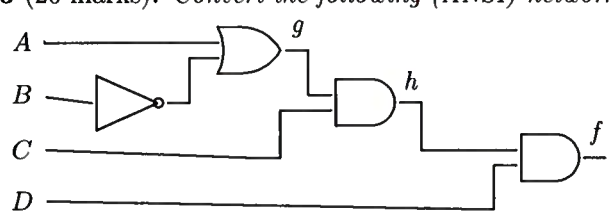
$k = \bar{h} = \overline{A\bar{B} + C} = \overline{(A\bar{B}+C)} \neq \cancel{\bar{A}B} + \bar{C}$

$\boxed{f = k \cdot D = \overline{(A\bar{B}+C)}D}$

Truth table

#	A	B	C	D	$\bar{B}$	$A\bar{B}$	$A\bar{B}+C$	$\overline{A\bar{B}+C}$
0	0	0	0	0	1			
1	0	0	0	1	1			
2	0	0	1	0	1			
3	0	0	1	1	1			
4	0	1	0	0	0			
5	0	1	0	1	0			
6	0	1	1	0	0			
7	0	1	1	1	0			
8	1	0	0	0	1			
9	1	0	0	1	1			
10	1	0	1	0	1			
11	1	0	1	1	1			
12	1	1	0	0	0			
13	1	1	0	1	0			
14	1	1	1	0	0			
15	1	1	1	1	0			

Problem 1.3 (20 marks). *Convert the following (ANSI) network into a Boolean expression and a truth table.*



$f = \underbrace{\underbrace{(A+\bar{B})}_{g} \cdot C}_{h} \cdot D$

A	B	C	D	f
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$\bar{B}CD$

ACD

Truth table

#	A	B	C	D	$\bar{B}$	$A\bar{B}$	$A\bar{B}+C$	$\overline{A\bar{B}+C}$	$\overline{(A\bar{B}+C)}\cdot D$
0	0	0	0	0	1	0	0	1	0
1	0	0	0	1	1	0	0	1	1
2	0	0	1	0	1	0	1	0	0
3	0	0	1	1	1	0	1	0	0
4	0	1	0	0	0	0	0	1	0
5	0	1	0	1	0	0	0	1	1
6	0	1	1	0	0	0	1	0	0
7	0	1	1	1	0	0	1	0	0
8	1	0	0	0	1	1	1	0	0
9	1	0	0	1	1	1	1	0	0
10	1	0	1	0	1	1	1	0	0
11	1	0	1	1	1	1	1	0	0
12	1	1	0	0	0	0	0	1	0
13	1	1	0	1	0	0	0	1	1
14	1	1	1	0	0	0	1	0	0
15	1	1	1	1	0	0	1	0	0

1.6 Minterms and Maxterms

1.6.1 Minterms

Minterm is a product involving all inputs (or complements) to a function. Every row of a truth table has a corresponding minterm. Minterm is true if and only if the corresponding row in the table is active.

Minterms defined as follows for each row of a two input truth table:

	A	B	minterm	minterm name
0	0	0	$\bar{A}\bar{B}$	m_0
1	0	1	$\bar{A}B$	m_1
2	1	0	$A\bar{B}$	m_2
3	1	1	AB	m_3

A	B	$\bar{A}B = m_1$	m_3	$m_1 + m_3$
0	0	0	0	0
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1

Consider a two input circuit whose output Y is given by the truth table:

A	B	Y	minterm	minterm name
0	0	0	$\bar{A}\bar{B}$	m_0
0	1	1	$\bar{A}B$	m_1
1	0	0	$A\bar{B}$	m_2
1	1	1	AB	m_3

then $Y = \bar{A}B + AB = m_1 + m_3 = \sum(1,3)$.
This also gives the *sum of products canonical form*.

m_1 $\quad$ A B

$1_{10} \longrightarrow 01 \longrightarrow \bar{A}B$ (m_1)

Example 1.6. *What is the minterm m_{13} for a 4-input circuit with inputs x, y, z, w (ordered from MSB to LSB)* $x\,y\,z\,w$

$(13)_{10} \longrightarrow (1101)_2$ ← LSB

MSB

$m_{13} = x\,y\,\bar{z}\,w$

MSB
Most Significant Bit

LSB
Least Significant Bit

Problem 1.4 (5 marks). *What is the minterm m_{23} for a 5-input circuit with inputs a, b, c, d, e (ordered from MSB to LSB).*

$m_{23} = a\bar{b}cde$

Example 1.7. *Convert the following 4-input truth table into sum of minterms and sum of products canonical form.*

$\boxed{ABC \neq \bar{A}BC}$

minterm name	*A*	*B*	*C*	*D*	*f*
m_0	0	0	0	0	0
m_1	0	0	0	1	1
m_2	0	0	1	0	0
m_3	0	0	1	1	0
m_4	0	1	0	0	0
m_5	0	1	0	1	1
m_6	0	1	1	0	0
m_7	0	1	1	1	0
m_8	1	0	0	0	0
m_9	1	0	0	1	0
m_{10}	1	0	1	0	0
m_{11}	1	0	1	1	0
m_{12}	1	1	0	0	0
m_{13}	1	1	0	1	1
m_{14}	1	1	1	0	0
m_{15}	1	1	1	1	0

Sum of minterms

$f = m_1 + m_5 + m_{13} = \Sigma(1,5,13)$

$= \Sigma m(1,5,13)$

Sum of products canonical form

$$\boxed{f = \bar{A}\bar{B}\bar{C}D + \underbrace{\bar{A}B\bar{C}D}_{m_5} + \underbrace{AB\bar{C}D}_{m_{13}}}$$

$5_{10} \rightarrow 0101$
ABCD

Problem 1.5 (10 marks). *Convert the following 4-input truth table into sum of minterms and sum of products canonical form.*

minterm name	*A*	*B*	*C*	*D*	*f*
m_0	0	0	0	0	0
m_1	0	0	0	1	0
m_2	0	0	1	0	0
m_3	0	0	1	1	1
m_4	0	1	0	0	0
m_5	0	1	0	1	0
m_6	0	1	1	0	0
m_7	0	1	1	1	1
m_8	1	0	0	0	0
m_9	1	0	0	1	0
m_{10}	1	0	1	0	0
m_{11}	1	0	1	1	1
m_{12}	1	1	0	0	0
m_{13}	1	1	0	1	1
m_{14}	1	1	1	0	1
m_{15}	1	1	1	1	0

$f = m_3 + m_7 + m_{11} + m_{13} + m_{15}$

sum of products canonical form:

$f = \bar{A}\bar{B}CD + \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABCD$

1.6.2 Maxterms (Dual of minterms)

Maxterm is a sum involving all inputs (or complements) to a function. Every row of a truth table has a corresponding maxterm. ~~Minterm~~ Maxterm is false if and only if the corresponding row in the table is active.

Maxterms are defined as follows for each row of a two input truth table:

A	B	maxterm	maxterm name
0	0	$A+B$	M_0
0	1	$A+\bar{B}$	M_1
1	0	$\bar{A}+B$	M_2
1	1	$\bar{A}+\bar{B}$	M_3

A	B	M_0 $A+B$	M_2 $\bar{A}+B$
0	0	0	1
0	1	1	1
1	0	1	0
1	1	1	1

Consider a two input circuit whose output Y is given by the truth table:

A	B	Y	maxterm	maxterm name
0	0	0	$A+B$	M_0
0	1	1	$A+\bar{B}$	M_1
1	0	0	$\bar{A}+B$	M_2
1	1	1	$\bar{A}+\bar{B}$	M_3

$M_2 = \bar{A} + B$

A B

$(2)_{10} \rightarrow (10)_2$

then $Y = (A+B)(\bar{A}+B) = M_0 M_2$.
Writing a functional specification in terms of minterms is also called product of sums canonical form.

Example 1.8. *Write the maxterm M_{11} for 4-input Boolean function with the ordered inputs A, B, C, D.*

MSB $\rightarrow$ A ... D $\leftarrow$ LSB $\quad (11)_{10} = (1011)_2 \quad M_{11} = \bar{A} + B + \bar{C} + \bar{D}$

Example 1.9. *Convert the following 4-input truth table into product of maxterms and product of sums canonical form.*

maxterm name	*A*	*B*	*C*	*D*	*f*
M_0	0	0	0	0	0
M_1	0	0	0	1	0
M_2	0	0	1	0	0
M_3	0	0	1	1	1
M_4	0	1	0	0	0
M_5	0	1	0	1	0
M_6	0	1	1	0	0
M_7	0	1	1	1	1
M_8	1	0	0	0	0
M_9	1	0	0	1	0
M_{10}	1	0	1	0	0
M_{11}	1	0	1	1	1
M_{12}	1	1	0	0	0
M_{13}	1	1	0	1	1
M_{14}	1	1	1	0	1
M_{15}	1	1	1	1	0

$f = M_0 M_1 M_2 M_4 M_5 M_6 M_8 M_9 M_{10} M_{12} M_{15}$

$= \prod M(0,1,2,4,5,6,8,9,10,12,15)$

$f = (A+B+C+D)(A+B+C+\bar{D})(A+B+\bar{C}+D)(A+\bar{B}+C+D)$
$(A+\bar{B}+C+\bar{D})(A+\bar{B}+\bar{C}+D)(\bar{A}+B+C+D)(\bar{A}+B+C+\bar{D})$
$(\bar{A}+B+\bar{C}+D)(\bar{A}+\bar{B}+C+D)(\bar{A}+\bar{B}+\bar{C}+\bar{D})$

Product of sums canonical form

Problem 1.6 (10 marks). *Convert the following 4-input truth table into product of maxterms and products of sums canonical form.*

maxterm name	A	B	C	D	f
M_0	0	0	0	0	0
M_1	0	0	0	1	1
M_2	0	0	1	0	1
M_3	0	0	1	1	1
M_4	0	1	0	0	1
M_5	0	1	0	1	0
M_6	0	1	1	0	1
M_7	0	1	1	1	1
M_8	1	0	0	0	0
M_9	1	0	0	1	1
M_{10}	1	0	1	0	1
M_{11}	1	0	1	1	1
M_{12}	1	1	0	0	0
M_{13}	1	1	0	1	1
M_{14}	1	1	1	0	1
M_{15}	1	1	1	1	0

Maxterm form

$f = M_0 \cdot M_5 \cdot M_8 \cdot M_{12} \cdot M_{15}$

~~Sum of~~ Product of sums canonical form

$f = (A+B+C+D) \cdot (A+\bar{B}+C+\bar{D}) \cdot (\bar{A}+B+C+D) \cdot (\bar{A}+\bar{B}+C+D) \cdot (\bar{A}+\bar{B}+\bar{C}+\bar{D})$

Example 1.10. *Write the 3-input truth table for the function $f = m_2 + m_3 + m_7$.*

$x\ y\ z$

#	x	y	z	f
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	0
6	1	1	0	0
7	1	1	1	1

Problem 1.7 (10 marks). *Write the 3-input truth table for the function $f = M_4 M_5 M_7$.*

x	y	z	f
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Problem 1.8 (10 marks). *Write the truth table for the function $f = \bar{A}B\bar{C} + AB\bar{C}$.*

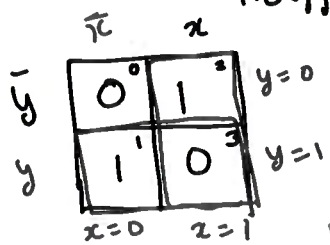
A	B	C	f
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

1.7 Karnaugh maps (Truth table + Venn Diagram)

Helpful in simplifying Boolean expressions
Digital circuits

Two input K-maps

	x	y	f
m_0	0	0	0
m_1	0	1	1
m_2	1	0	1
m_3	1	1	0

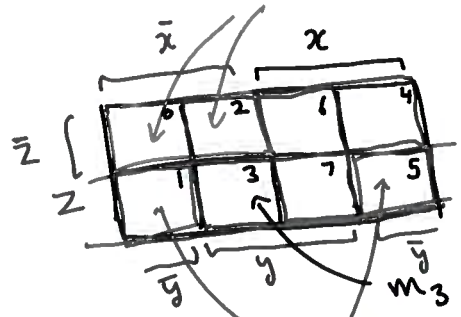


$m_0 + m_2 = \bar{x}\cdot\bar{z}$

Three input K-maps

MSB → LSB

x	y	z	f
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	



Each rectangular grouping of 2, 4, 8, 16, ... corresponds to a product term

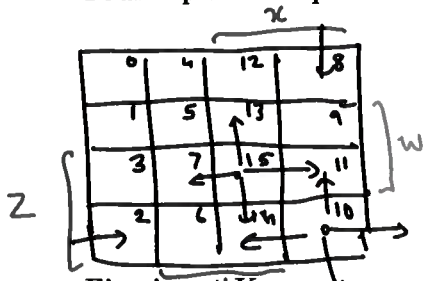
$m_3 = \bar{x}yz$

$m_1 + m_5 = z\cdot\bar{y}$

$\bar{x}\bar{y}z + x\bar{y}z = (\bar{x}+x)(\bar{y}z) = 1\cdot\bar{y}z$

Four input K-maps

MSB → LSB
$x\ y\ z\ w$

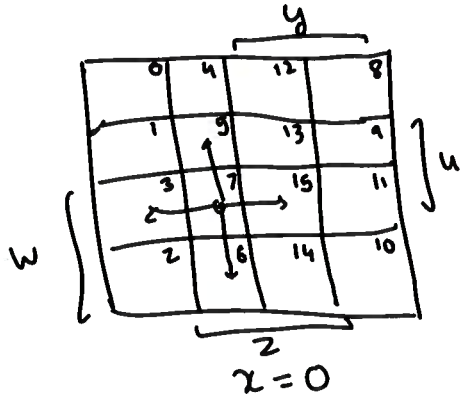


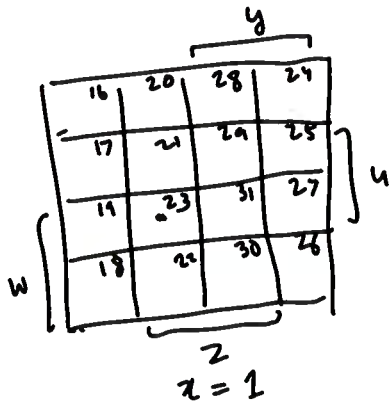
4 input k-map must have 4-neighbors for each cell

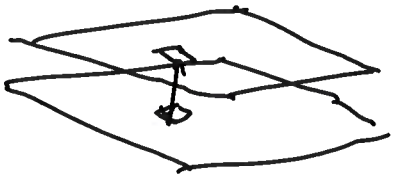
K-map wraps around

Five input K-maps

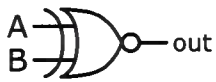
MSB → LSB
$x\ y\ z\ w\ u$







1.8 More Gates and notations summary

Name	C/Verilog	Boolean expr.	Truth Table	(ANSI) symbol	K-map
NAND Gate	`Q = ~(x1 & x2)`	$Q = \overline{x_1 \cdot x_2} = \overline{x_1 x_2}$	x_1 x_2 $\overline{x_1 \cdot x_2}$ 0 0 1 0 1 1 1 0 1 1 1 0		A: 0 1; B: 0: 1 1; 1: 1 0
NOR Gate	`Q = ~(x1 | x2)`	$Q = \overline{x_1 + x_2}$	x_1 x_2 $\overline{x_1 + x_2}$ 0 0 1 0 1 0 1 0 0 1 1 0		A: 0 1; B: 0: 1 0; 1: 0 0
XOR Gate	`Q = x1 ^ x2`	$Q = x_1 \oplus x_2$	x_1 x_2 $x_1 \oplus x_2$ 0 0 0 0 1 1 1 0 1 1 1 0		A: 0 1; B: 0: 0 1; 1: 1 0
XNOR Gate	`Q = ~(x1 ^ x2)`	$Q = \overline{x_1 \oplus x_2}$	x_1 x_2 $\overline{x_1 \oplus x_2}$ 0 0 1 0 1 0 1 0 0 1 1 1		A: 0 1; B: 0: 1 0; 1: 0 1

XOR = Exclusive OR gate

x	y	f
0	0	0
1	0	1
0	1	1
1	1	0

$f = x \oplus y$

$f = x \wedge y$ in C/verilog

Example 1.11. *Convert the following Boolean expression into a K-map.* $f = (\overline{A\bar{B} + C})D$

MSB → LSB

A B C D

4-input K map

$A\bar{B}+C$

0	0	0	1
0	0	0	1
1	1	1	1
1	1	1	1

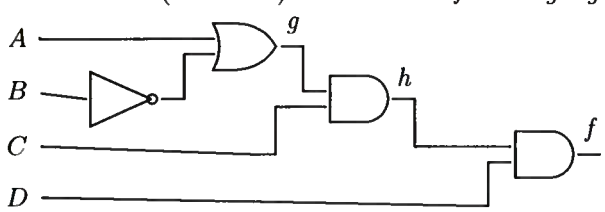
A; D; C; $\bar{B}$ B $\bar{B}$

$f = (\overline{A\bar{B}+C})D$

0	0	0	0
1	1	1	0
0	0	0	0
0	0	0	0

A; D; C; $\bar{B}$ B $\bar{B}$

Problem 1.9 (10 marks). *Convert the following logic circuit into a K-map.*



$f = (A+\bar{B}) \cdot C \cdot D$

0	0	0	0
0	0	0	0
1	0	1	1
0	0	0	0

A; D; B; C; B

1.9 Boolean Algebra

1.9.1 Axioms of Boolean algebra

1. $0 \cdot 0 = 0$
2. $1 + 1 = 1$

Definitions

≠ Axioms are fundamental truths that you take to be true and do not ask proofs for.